B. Tenzing and Books

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: bitmasks, greedy, math | https://codeforces.com/problemset/problem/1842/B

Tenzing received 3*n* books from his fans. The books are arranged in 3 stacks with *n* books in each stack. Each book has a non-negative integer difficulty rating.

Tenzing wants to read some (possibly zero) books. At first, his *knowledge* is 0.

To read the books, Tenzing will choose a non-empty stack, read the book on the top of the stack, and then discard the book. If Tenzing's knowledge is currently *u*, then his knowledge will become $u|v$ after reading a book with difficulty rating *v*. Here | denotes the bitwise OR operation. Note that Tenzing can stop reading books whenever he wants.

Tenzing's favourite number is *x*. Can you help Tenzing check if it is possible for his knowledge to become *x*?

Input

Each test contains multiple test cases. The first line of input contains a single integer t ($1 \le t \le 10^4$) — the number of test cases. The description of test cases follows.

The first line of each test case contains two integers n and x ($1 \le n \le 10^5$, $0 \le x \le 10^9$) — the number of books in each stack and Tenzing's favourite number.

The second line of each test case contains n integers $a_1,a_2,\ldots,a_n$ ($0 \le a_i \le 10^9$) — the difficulty rating of the books in the first stack, from top to bottom.

The third line of each test case contains n integers $b_1,b_2,\ldots,b_n$ ($0 \le b_i \le 10^9$) — the difficulty rating of the books in the second stack, from top to bottom.

The fourth line of each test case contains n integers $c_1,c_2,\ldots,c_n$ ($0 \le c_i \le 10^9$) — the difficulty rating of the books in the third stack, from top to bottom.

It is guaranteed that the sum of n does not exceed 10^5.

Output

For each test case, output "`Yes`" (without quotes) if Tenzing can make his knowledge equal to *x*, and "`No`" (without quotes) otherwise.

You can output the answer in any case (upper or lower). For example, the strings "`yEs`", "`yes`", "`Yes`", and "YES" will be recognized as positive responses.

Example

Input

```
3
5 7
1 2 3 4 5
5 4 3 2 1
1 3 5 7 9
5 2
3 2 3 4 5
5 4 3 2 1
3 3 5 7 9
3 0
1 2 3
3 2 1
2 2 2
```

Output

```
Yes
No
Yes
```

Note

For the first test case, Tenzing can read the following 4 books:

- read the book with difficulty rating 1 on the top of the first stack. Tenzing's knowledge changes to 0|1=1.
- read the book with difficulty rating 1 on the top of the third stack. Tenzing's knowledge changes to 1|1=1.
- read the book with difficulty rating 2 on the top of the first stack. Tenzing's knowledge changes to 1|2=3.
- read the book with difficulty rating 5 on the top of the second stack. Tenzing's knowledge changes to 3|5=7.

After reading all books, Tenzing's knowledge is 7.

For the third test case, Tenzing can read 0 books to make his final knowledge equals to 0.